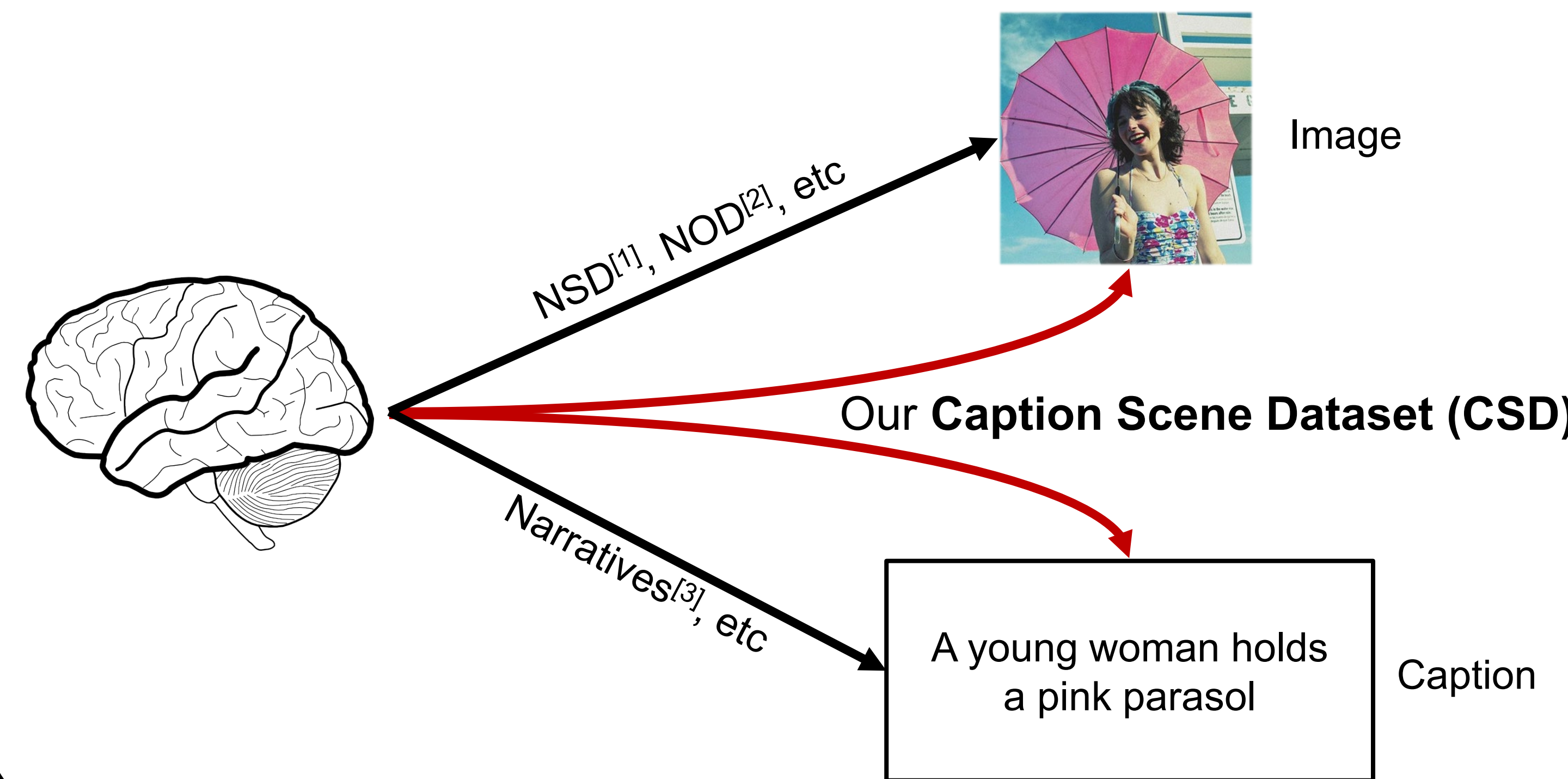


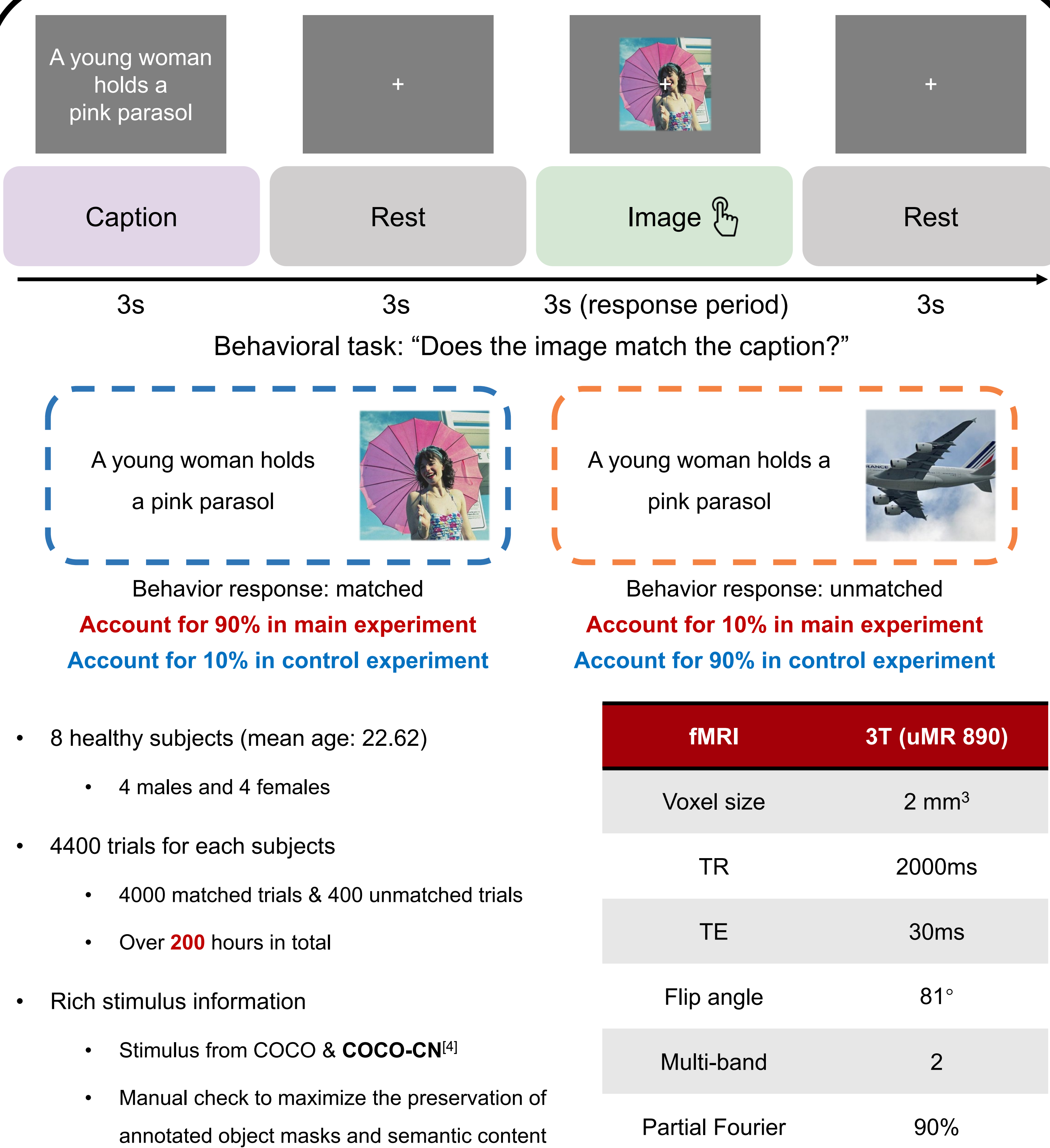


Introduction

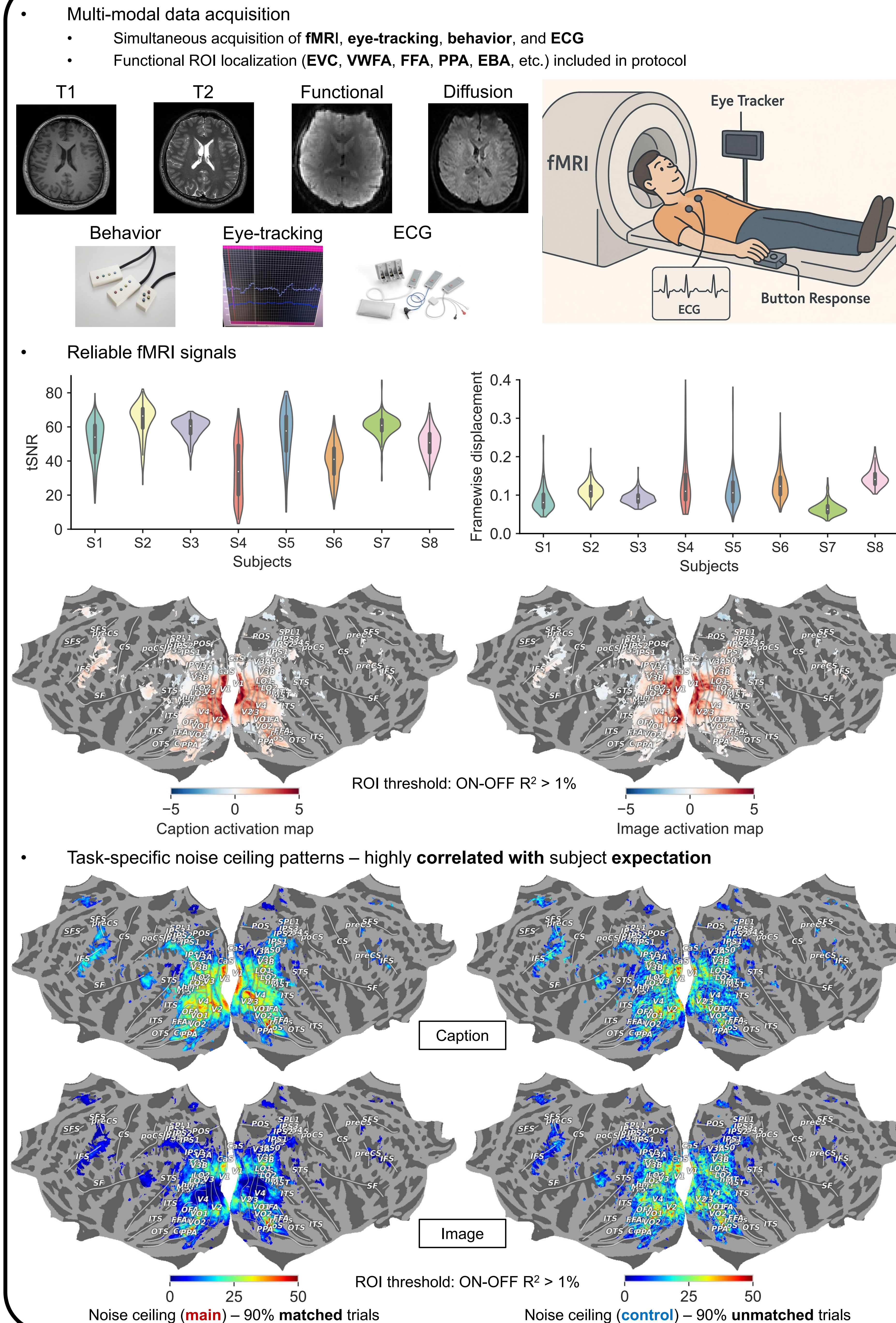
- Large-scale fMRI datasets enable AI-based modeling of brain function
- Most datasets are unimodal—limiting cross-modal semantic research
- Our study focuses on **vision-language semantic matching** using naturalistic stimuli.



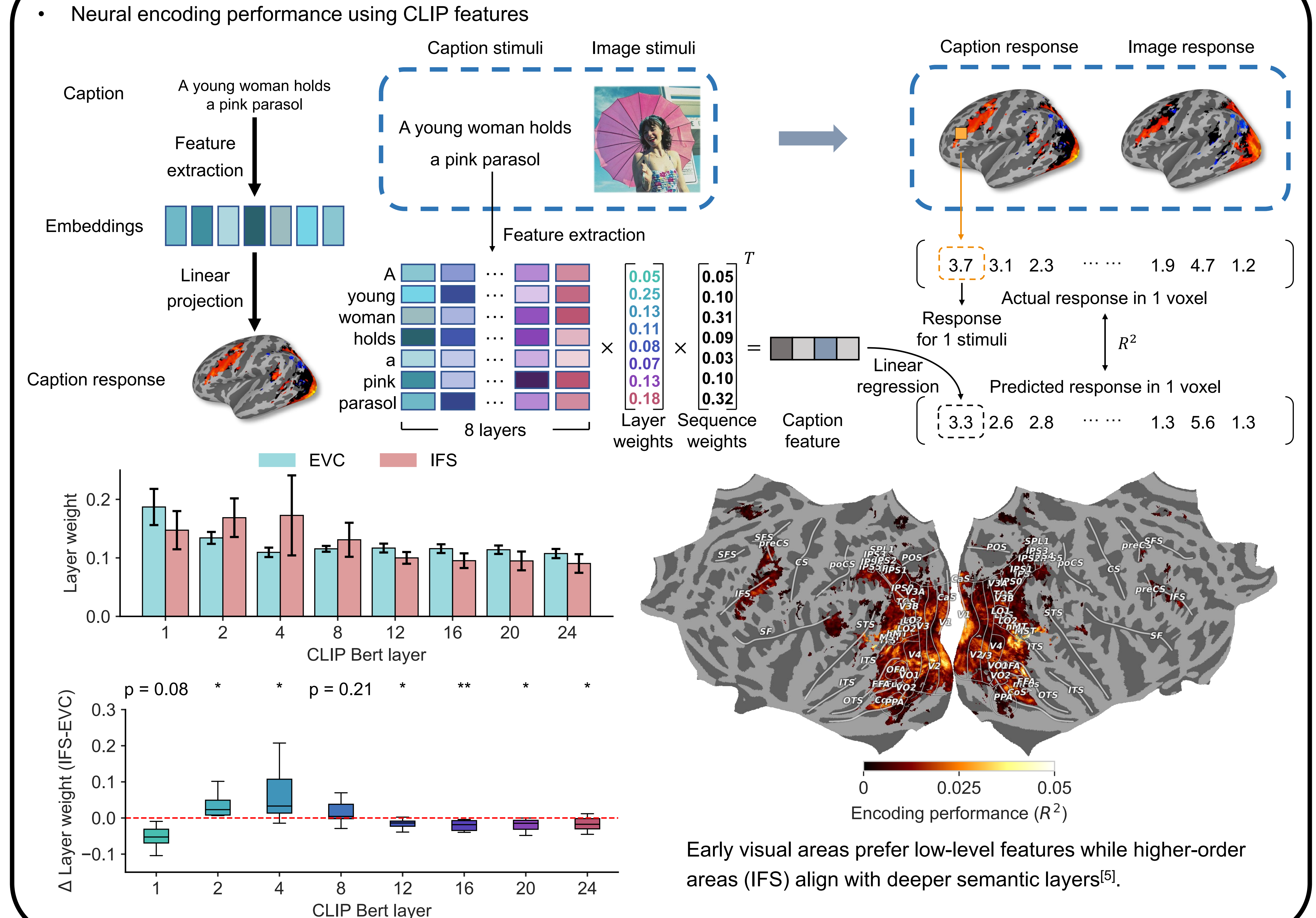
Experiment Design



Dataset Features



Neural Encoding Model



Take-home Message

- We present **Caption Scene Dataset, CSD**, a large-scale, cross-modal fMRI dataset designed to **investigate semantic processing** across vision and language
- Data reliability** is validated through quality metrics (tSNR, motion) and **proposed encoding models** that replicate known neural feature hierarchies

References

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